2025 SSMO Speed Round Solutions

SMO Team

SR 1: Define

$$f(x) = \begin{cases} x+2 & \text{if } x < 0 \\ x^2+1 & \text{if } x \geq 0 \end{cases}$$

and let $x_0 = -3$. Define $x_{n+1} = f(x_n)$. Find the least n such that $x_n > 100$.

Answer: $\boxed{6}$

Solution: We compute the first few values of x_n

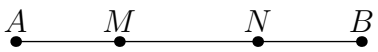
$$\begin{aligned} x_0 &= -3 \\ x_1 &= f(-3) = -1 \\ x_2 &= f(-1) = 1 \\ x_3 &= f(1) = 2 \\ x_4 &= f(2) = 5 \\ x_5 &= f(5) = 26 \\ x_6 &= f(26) = 677. \end{aligned}$$

Thus, the smallest n such that $x_n > 100$ is $\boxed{6}$.

SR 2: Let A and B be points such that $AB = 50$. Points M and N lie on $\overline{AB}$ such that M lies between points A and N and N lies between points B and M. Given that $MN = 20$ and $AN \cdot BM$ is maximized, find the length of AM.

Answer: $\boxed{15}$

Solution:



Note that $AN + BM = AB + MN = 70$. By AM-GM, the value of $AN \cdot BM$ is at a maximum when $AN = BM = 35$. Thus, $AM = AN - MN = \boxed{15}$.

SR 3: Anna is buying different types of cheese from the local supermarket. Let x, y, and z be the number of pieces of blue, cheddar, and mozzarella cheese, respectively, that Anna buys. She can buy any nonnegative integer number of each type, but the total number of pieces must be at most 12. How many different combinations (x, y, z) of cheese can Anna buy? (Anna is allowed to buy 0 pieces of cheese.)

Answer: $\boxed{455}$

Solution: We seek the number of ordered triples (x, y, z) of nonnegative integers such that $0 \le x + y + z \le 12$. Let $0 \le n \le 12$ be some integer. By stars and bars, there are $\binom{n+2}{2}$ ordered triples (x, y, z) of nonnegative integers satisfying $x + y + z = n$. Thus, the answer is the sum

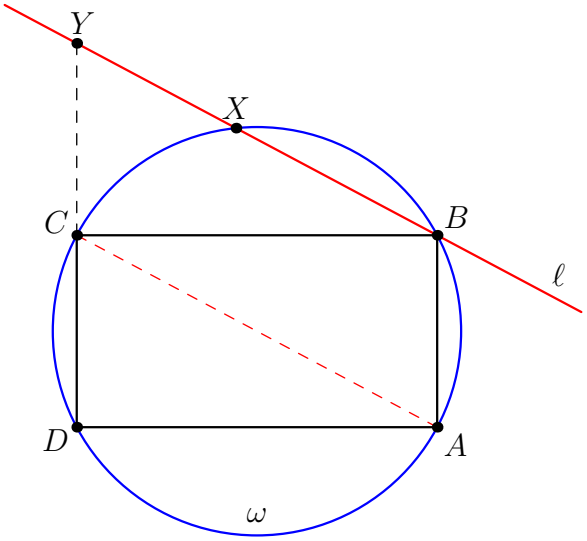
$$\binom{2}{2} + \binom{3}{2} + \cdots + \binom{14}{2}.$$

By the hockey stick identity, this is equal to $\binom{15}{3} = \boxed{455}$.

SR 4: In rectangle $ABCD$, let $AB = 8$, $BC = 15$, ω be the circumcircle of $ABCD$, ℓ be the line through B parallel to AC, and $X \neq B$ be the intersection of ℓ and ω. The length BX can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{178}$

Solution:



Extend CD to intersect ℓ at point Y. Note that $AB \parallel CY$ and $AC \parallel BY$, so $ABYC$ is a parallelogram. This implies that $BY = AC = \sqrt{8^2 + 15^2} = 17$ and $CY = AB = 8$. Let $t = BX$. By power of a point on Y with respect to ω, we have $XY \cdot BY = CY \cdot DY$. We know that $CY = 8$, $DY = 16$, $BY = 17$, and $XY = 17 - t$. Thus,

$$17(17 - t) = 128 \implies t = \frac{161}{17} \implies 161 + 17 = \boxed{178}.$$

SR 5: Let $N = 101112 \cdots 9899$ be the number formed when all the two-digit positive integers are concatenated in increasing order. How many ordered triples of digits (a, b, c) are there such that a, b, and c appear as consecutive digits (in that order) in the decimal representation of N?

Answer: $\boxed{170}$

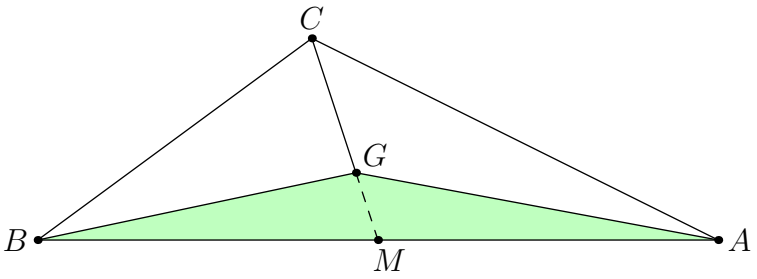
Solution: Since there are 90 two-digit positive integers, N has 180 digits. Then, there are $180 - 2 = 178$ (not necessarily distinct) ordered triples of digits (a, b, c) such that the string of digits abc appears in the decimal representation of N. Furthermore, any such string abc can appear at most twice in N. This is because if abc appears in N, then either $\overline{ab}$ or $\overline{bc}$ is one of the two-digit positive integers that was originally concatenated to form

N. Each of these two possibilities fixes the position of the string abc among the digits of N because every two-digit positive integer was used exactly once in forming N. Therefore, we will count the number of strings abc that appear twice in N and then subtract that number from 178. Suppose the string abc appears twice in N. Let ? denote any unknown digit. Then, $\overline{ab}$ and $\overline{c?}$ are consecutive two-digit positive integers in that order, and $\overline{?a}$ and $\overline{bc}$ are also consecutive two-digit positive integers in that order. If $\overline{?a}$ and $\overline{bc}$ are consecutive, we must have $a \neq c$. If $\overline{ab}$ and $\overline{c?}$ are consecutive in that order and $a \neq c$, we must have $c = a + 1$ and $b = 9$. This implies that $(a, b, c) = (d, 9, d + 1)$, where d is some digit. Now, the string abc appears twice in N so long as each of $\overline{ab}$, $\overline{c?}$, $\overline{?a}$, and $\overline{bc}$ is indeed a two-digit positive integer. This is equivalent to a, b, and c all being nonzero digits, which means that we must have $d \in \{1, 2, \ldots, 8\}$. Each such value of d corresponds to a unique string abc that appears twice in N, so the answer is $178 - 8 = \boxed{170}$.

SR 6: The centroid G of $\triangle ABC$ has distances 21, 60, and 28 from sides AB, BC, and CA, respectively. Find the perimeter of $\triangle ABC$.

Answer: $\boxed{630}$

Solution:



Let $a = BC$, $b = CA$, and $c = AB$. If M denotes the midpoint of AB, note that by centroid properties, $GM : CM = 1 : 3$, and hence, $[ABG] = [BGM] + [AGM] = \frac{1}{3}[BCM] + \frac{1}{3}[ACM] = \frac{1}{3}[ABC]$ by same-altitude triangles. Using analogous reasoning, we can show that $[ABG] = [BCG] = [CAG] = \frac{1}{3}[ABC]$. We know that the length of the altitude from G to AB is 21, and thus $[ABG] = \frac{21}{2}c$. Similarly, we can find that $[BCG] = 30a$ and $[CAG] = 14b$. Equating the areas of these triangles, we have $30a = 14b = \frac{21}{2}c$, which is equivalent to $a : b : c = 7 : 15 : 20$. Let $a = 7x$, $b = 15x$, and $c = 20x$ for some positive real x; we now solve for x by computing the area of ABC in two different ways. On one hand, we have $[ABC] = [ABG] + [BCG] + [CAG] = 630x$. On the other hand, by Heron's formula, $[ABC] = 42x^2$. Hence, $630x = 42x^2$, which gives $x = 15$. The perimeter of ABC is $42x = \boxed{630}$.

SR 7: Positive integers a and b satisfy $63a = 40b$. The sum of all possible values of $\frac{\varphi(a)}{\varphi(b)}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{72}$

Solution: Let $a = 2^{w+3}3^x5^{y+1}7^zT$ and $b = 2^w3^{x+2}5^y7^{z+1}T$, where w, x, y, and z are all nonnegative integers and T is a positive integer not divisible by any of 2, 3, 5, and 7. Then,

$$\frac{\varphi(a)}{\varphi(b)} = \frac{\varphi(2^{w+3})}{\varphi(2^w)} \cdot \frac{\varphi(3^x)}{\varphi(3^{x+2})} \cdot \frac{\varphi(5^{y+1})}{\varphi(5^y)} \cdot \frac{\varphi(7^z)}{\varphi(7^{z+1})}.$$

Note that $\frac{\varphi(2^{w+3})}{\varphi(2^w)}$ is 4 if $w = 0$ and 8 if $w > 0$. Similarly, $\frac{\varphi(3^x)}{\varphi(3^{x+2})}$ is $\frac{1}{6}$ if $x = 0$ and $\frac{1}{9}$ if $x > 0$, $\frac{\varphi(5^{y+1})}{\varphi(5^y)}$ is 4 if $y = 0$ and 5 if $y > 0$, and $\frac{\varphi(7^z)}{\varphi(7^{z+1})}$ is $\frac{1}{6}$ if $z = 0$ and $\frac{1}{7}$ if $z > 0$. The resulting $2^4 = 16$ products are distinct, since the exponents of 5, 7, 3, and 2 in the product determine the choices for y, z, x, and w, in that order. So, the sum of all possible values of $\frac{\varphi(a)}{\varphi(b)}$ is

$$(4+8)\left(\frac{1}{6}+\frac{1}{9}\right)(4+5)\left(\frac{1}{6}+\frac{1}{7}\right) = \frac{65}{7} \implies 65+7 = \boxed{72}.$$

SR 8: Let S be the set of all ordered pairs (P, Q), where P and Q are subsets of $\{1, 2, \ldots, 25\}$ satisfying $|P \cup Q| = 17$ and $|(P \cap Q) \cap \{20, 25\}| \geq 1$. If an ordered pair (A, B) is chosen uniformly at random from S, the expected value of $|A \cap B|$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{29}$

Solution: For each $i \in \{1, 2, \ldots, 25\}$, let p_i denote the probability that $i \in A \cap B$. Since $|A \cap B|$ counts the elements i with $i \in A \cap B$, from linearity of expectation, we have

$$\mathbb{E}(|A \cap B|) = p_1 + p_2 + \cdots + p_{25} = 23p_1 + 2p_{20},$$

since $p_i = p_1$ for all $i \neq 20, 25$ and $p_{25} = p_{20}$ by symmetry. Note that each element of $A \cup B$ lies in A only, in B only, or in both. If 20 and 25 are both in $A \cup B$, there are 5 ways to place them, 3 of which have $20 \in A \cap B$, and $3^{15}\binom{23}{15}$ ways to choose and place the other 15 elements of $A \cup B$. Otherwise, one of 20 and 25 lies in $A \cap B$ and the other lies in neither set, so there are 2 ways to place them and $3^{16}\binom{23}{16}$ ways to choose and place the other 16 elements. Since

$$2 \cdot 3^{16}\binom{23}{16} = 2 \cdot 3^{16} \cdot \frac{8}{16}\binom{23}{15} = 3 \cdot 3^{15}\binom{23}{15},$$

the first case occurs with probability $\frac{5}{8}$ and the second with probability $\frac{3}{8}$. So,

$$p_{20} = \frac{5}{8} \cdot \frac{3}{5} + \frac{3}{8} \cdot \frac{1}{2} = \frac{9}{16}.$$

In the two cases, 1 lies in $A \cup B$ with probability $\frac{15}{23}$ and $\frac{16}{23}$, respectively, and then lies in $A \cap B$ with probability $\frac{1}{3}$. So,

$$p_1 = \frac{1}{3}\left(\frac{5}{8} \cdot \frac{15}{23} + \frac{3}{8} \cdot \frac{16}{23}\right) = \frac{41}{184}.$$

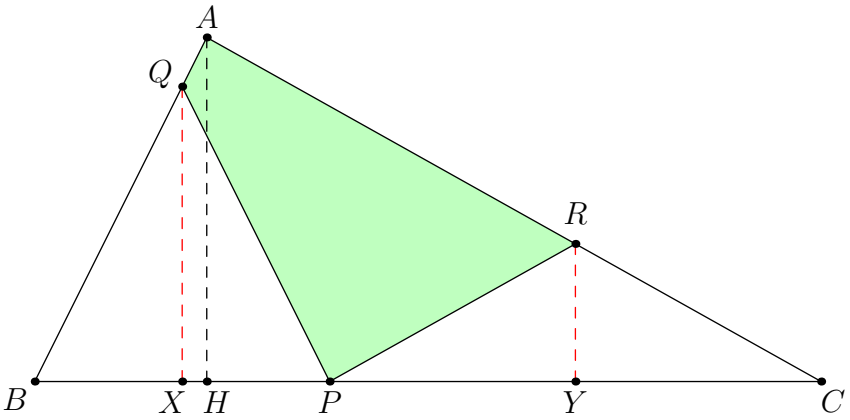
Thus,

$$\mathbb{E}(|A \cap B|) = 23 \cdot \frac{41}{184} + 2 \cdot \frac{9}{16} = \frac{25}{4} \implies 25 + 4 = \boxed{29}.$$

SR 9: Let ABC be a triangle. The point P lies on side BC, the point Q lies on side AB, and the point R lies on side AC such that $PQ = BQ$, $CR = PR$, and $\angle APB < 90°$. Let H be the foot of the altitude from A to BC. Given that $BP = 3$, $CP = 5$, and $[AQPR] = \frac{3}{7} \cdot [ABC]$, the value of $BH \cdot CH$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{191}$

Solution:



Let X and Y be the projections of Q and R onto BC, respectively. Since $\triangle BQP$ and $\triangle CRP$ are both isosceles, X and Y are the midpoints of BP and CP, respectively. Let $BH = x$, $CH = y$, and $AH = h$. Note that $x + y = BC = 8$, and also note that the condition $\angle APB < 90^\circ$ implies $x = BH < BP = 3$. Next, note that $\triangle QBX \sim \triangle ABH$, and hence $\frac{QX}{BX} = \frac{AH}{BH}$. From this, we get $QX = \frac{AH \cdot BX}{BH} = \frac{3h}{2x}$. Analogously, from the similarity $\triangle RCY \sim \triangle ACH$, we get $RY = \frac{5h}{2y}$. Thus,

$$[BQP] + [CRP] = \frac{9h}{4x} + \frac{25h}{4y}.$$

Now, using the area condition in the problem statement, we have

$$\begin{aligned} \frac{9h}{4x} + \frac{25h}{4y} &= [BQP] + [CRP] \\ &= [ABC] - [AQPR] \\ &= \frac{4}{7}[ABC] \\ &= \frac{16h}{7}. \end{aligned}$$

Cancelling a factor of h from the first and last equations in the above chain and substituting $y = 8 - x$, we obtain

$$\frac{9}{4x} + \frac{25}{4(8-x)} = \frac{16}{7}.$$

Solving this equation, we find $x = \frac{7}{4}, \frac{9}{2}$. Since $x < 3$, we must have $x = \frac{7}{4}$, which implies $y = \frac{25}{4}$. Hence,

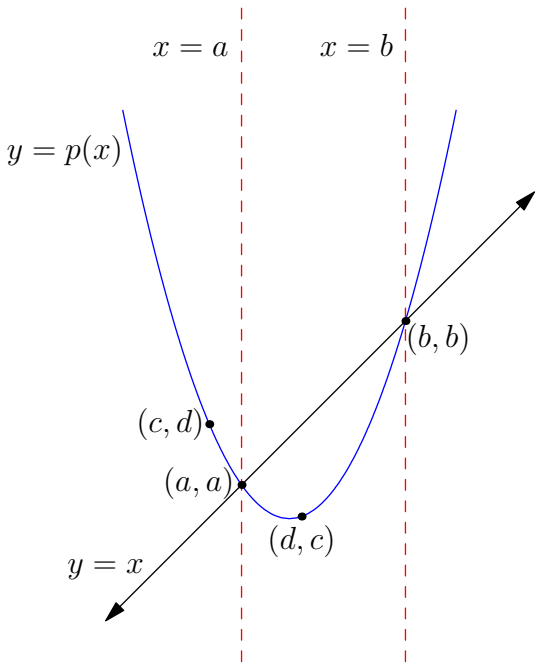
$$BH \cdot CH = xy = \frac{175}{16} \implies 175 + 16 = \boxed{191}.$$

SR 10: Let p be a quadratic with a positive leading coefficient, and let r be a real number satisfying $r < 1 < \frac{5}{2r} < 5$. Given that $p(p(x)) = x$ for $x \in \{r, 1, \frac{5}{2r}, 5\}$, find $p(12)$.

Answer: $\boxed{166}$

Solution: Note that any fixed point of $p(x)$ is also a fixed point of $p(p(x))$, and the four roots of the quartic $p(p(x)) - x$ are r, 1, $\frac{5}{2r}$, and 5. So, the fixed points of p are a subset of $\{r, 1, \frac{5}{2r}, 5\}$. If $p(c) = d \neq c$ for some c in this set, then $p(d) = p(p(c)) = c$, so the elements of the set that are not fixed points of p pair up, meaning p has 0, 2, or 4 fixed points. We claim that p has exactly 2 fixed points, and that these fixed points are 1 and 5. First,

we prove that p has exactly two fixed points. It is impossible that p has 4 fixed points because that would imply $\deg p \geq 4$. It is also impossible that p has 0 fixed points. For the sake of contradiction, suppose that this is the case. Then, the parabola $y = p(x)$ does not intersect the line $y = x$, and since p has a positive leading coefficient, the parabola $y = p(x)$ lies entirely above the line $y = x$. Since p has no fixed points, the points $(1, p(1))$ and $(p(1), p(p(1))) = (p(1), 1)$ are distinct and lie on opposite sides of the line $y = x$. However, both of these points lie on the parabola $y = p(x)$, which is a contradiction. Thus, p has exactly 2 fixed points.



Now, we show that the fixed points of p are 1 and 5. Let (a, b, c, d) be a permutation of $(r, 1, \frac{5}{2r}, 5)$, and suppose that $p(a) = a$, $p(b) = b$, $p(c) = d$, and $p(d) = c$. WLOG, suppose $a < b$. We first note that a point on $y = p(x)$ lies under the line $y = x$ if and only if it lies between the lines $x = a$ and $x = b$. WLOG, suppose $a < d < b$. Since (d, c) lies under the line $y = x$, we have $c < d$. Note that (c, d) cannot lie between $x = a$ and $x = b$ because exactly one of the points (c, d) and (d, c) lies below $y = x$. Since $c < d$, it follows that $c < a < d < b$, so $(a, b, c, d) = (1, 5, r, \frac{5}{2r})$. Thus, $p(1) = 1$ and $p(5) = 5$. Let m be the leading coefficient of p, so $p(x) = m(x - 1)(x - 5) + x$. For convenience, replace r and $\frac{5}{2r}$ by s and t, respectively, satisfying $st = \frac{5}{2}$. We have

$$t = ms^2 - (6m - 1)s + 5m$$
$$s = mt^2 - (6m - 1)t + 5m.$$

Adding these two equations gives us $6m(s + t) = m(s^2 + t^2) + 10m$. Dividing through by m and using $s^2 + t^2 = (s + t)^2 - 2st = (s + t)^2 - 5$, we find that $s + t = 1$ or $s + t = 5$. Since $t > 1$ and $s > 0$, we must have $s + t = 5$. Then, subtracting the second equation from the first and rearranging gives us $t - s + m(t^2 - s^2) = (6m - 1)(t - s)$. As $s \neq t$, we may divide through by $t - s$ to obtain $1 + m(s + t) = 1 + 5m = 6m - 1$. Finally, we have $m = 2$, which gives $p(x) = 2x^2 - 11x + 10$. The answer is $2(12)^2 - 11(12) + 10 = \boxed{166}$.